Smart Voltage and Over-Current Protector (Using PIC18F4550 MCU)

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1. Objective

Design a microcontroller-based device that protects an electrical appliance from grid over voltage, under voltage and over current faults by disconnecting it through a relay before damage occurs. The device must support computer communication, and its settings must be changeable both from the device itself and from a computer.

2. System Design

The controller is a **PIC18F4550** (20 MHz crystal, 5 V supply). Mains voltage is sensed by **ZMPT101B** module and load current by **ACS712-20A** sensor , both read by the 10-bit ADC (AN0, AN1). A 5 V relay driven by a transistor with a flyback diode disconnects the load. An input MOV and a fuse protect against fast surges and short circuits. Low-voltage power comes from an isolated HLK-PM01 supply. A 16x2 LCD, three push buttons on individual input pins, a buzzer and two LEDs (normal / fault) provide local feedback. The PC connects through the on-chip **hardware UART (EUSART, 9600 baud)** and a USB-UART adapter (which appears as a virtual COM port on the PC. If we use wi-fi module that connect with our home wi-fi router and send data to the mobile app. Then we do not need to connect USB with the pc.

3. Firmware Operation

- The ADC is sampled every 200 us (100 samples per 20 ms mains cycle, placed by Timer1); RMS voltage and current are computed each cycle, together with peak values. Timer0 gives a 1 ms time base and the UART runs on interrupts.
- **Faults detected:** over-voltage, under-voltage, over-current, peak surge and short circuit.
- Normal faults trip after a user-set delay to avoid nuisance trips. Peak surge and short circuit trip immediately.
- State machine: **NORMAL -> TRIPPED -> WAIT_RECONNECT -> NORMAL, with LOCKOUT** after repeated trips (manual reset required). Hysteresis prevents relay chatter.
- Fail-safe start: the relay is OFF at power-up and turns ON only after settings are validated and voltage is normal. Watchdog timer and brown-out reset are enabled.

4. Limitations

- A relay opens in several milliseconds, so it cannot stop microsecond spikes. That job belongs to the MOV; the relay protects against sustained over-voltage and over-current.
- Low-cost sensor modules give roughly +/-2-3% accuracy; calibration against a multimeter is required.
- Native USB CDC was not implemented because it needs the Microchip USB stack and a PLL clock; a USB-UART adapter is used instead.
- Firmware has not been verified on real hardware unless Section 5 says otherwise; it was only syntax-checked on a PC.
- EEPROM writes (SAVE) take a few milliseconds per byte and briefly pause sampling.
- The PIC18F4550 has only 256 B of EEPROM, enough for settings but not for a long fault log.

5. Conclusion

The design uses the PIC18F4550's hardware UART, large flash (32 KB) and 2 KB RAM to detect voltage and current faults, disconnect the load, and allow configuration from both the device and a computer. Items listed in Section 6 as incomplete are future work.

6. References

1. Performance Analysis of a Microcontroller-Based Overvoltage and Overcurrent Protection System(https://alambiblio.com/ojs/index.php/ajoeee/article/view/77/58)